A Neural Network Based on Consistency Learning and Adversarial Learning for Semisupervised Synthetic Aperture Radar Ship Detection

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***Abstract*— Ship detection in synthetic aperture radar (SAR) images has important application value. Sea clutter, complex scenes, a large size change in ships, and the arbitrary directionality of ships make ship detection challenging. With the development of deep learning, many deep learning algorithms have been applied to SAR images. These algorithms need a lot of labeled data for training. It is time-consuming to label SAR data, and the unlabeled data are easy to obtain. It is necessary to use the unlabeled data effectively to improve the performance of the algorithm. In this study, a semisupervised SAR ship detection network, named the semisupervised consistency learning adversarial network (SCLANet), is presented. SCLANet is a two-stage detection network. The local features around the ship can be extracted by the SCLANet, and the features generated from unlabeled data become closer to those generated from labeled data by using adversarial learning. There are two consistency learning modules in SCLANet: noise robustness consistency learning and output encoding consistency learning. Noise robustness consistency learning can increase the robustness of the SCLANet. Maintaining consistency between the noisy results and the original results can train the unlabeled data. In output encoding consistency learning, outputs are mapped to a picture that is fed into an encoder to obtain the intermediate representation embedding. Another embedding is a layer in the main network of the SCLANet. Reducing the error between two embeddings can train the SCLANet with unlabeled data. Two types of consistency learning can be used as pretext tasks for semisupervised learning. Experiments were conducted on two SAR ship datasets. Compared with other algorithms, the SCLANet achieved the highest detection accuracy, indicating that it is more advantageous to use in ship detection.**

Manuscript received August 5, 2021; revised November 28, 2021 and December 20, 2021; accepted January 7, 2022. Date of publication January 11, 2022; date of current version March 11, 2022. This work was supported in part by the Key Scientific Technological Innovation Research Project by the Ministry of Education; in part by the National Natural Science Foundation of China under Grant 62171347, Grant 61877066, Grant 61771379, Grant 62001355, and Grant 62101405; in part by the Foundation for Innovative Research Groups of the National Natural Science Foundation of China under Grant 61621005; in part by the Key Research and Development Program in Shaanxi Province of China under Grant 2019ZDLGY03-05 and Grant 2021ZDLGY02-08; in part by the Science and Technology Program in Xi'an of China under Grant XA2020-RGZNTJ-0021; and in part by the 111 Project. *(Corresponding author: Biao Hou.)*

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Digital Object Identifier 10.1109/TGRS.2022.3142017

***Index Terms*— Convolutional neural network (CNN), object detection, semisupervised learning, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) is an active Earth observation system that can produce high-resolution radar images. SAR can be installed on airplanes, satellites, and spaceships to observe the Earth for 24 h without being affected by the weather [1]. Thus, the SAR system has unique advantages in disaster monitoring, urban planning, marine monitoring, agricultural planting, geographic mapping, and military application [2]–[8]. With the development of science and technology, many countries have developed their own SAR systems, such as Lacrosse, ERS, TerraSar, Gaofen-3, and Sentinel [9]. The current SAR system can obtain a large number of high-resolution SAR images. The representation of objects in an SAR image is not intuitive and is quite different from that in an optical image. Thus, the interpretation of an SAR image is challenging. Speckle and overlay often appear in SAR images, causing researchers to spend a lot of energy on image interpretation. To better interpret SAR images, a variety of algorithms have been designed.

SAR ship detection, which has increasingly received the attention of researchers, has important value in marine transportation and the military field. Due to the differences in radar signal reflection characteristics between a ship and seawater, a ship appears as a bright target, and seawater appears as a dark background in an SAR image [10]. Although there are obvious brightness differences between ships and the sea, ship wake and sea clutter affect the detection of ships. Given these characteristics, researchers have designed many traditional algorithms, including polarization-based algorithms, contrast-based algorithms, and statistical model-based constant false alarm rate (CFAR) algorithms [11]–[15]. However, the complex scene of an inshore area brings a lot of difficulties to ship detection, which makes traditional algorithms ineffective. Moreover, deep learning algorithms have been gradually applied to SAR images. Deep learning algorithms offer great innovation in the field of computer vision. Various algorithms based on the convolutional neural network (CNN) have greatly improved the accuracy of object detection. These object detection algorithms are generally divided into one-stage algorithms

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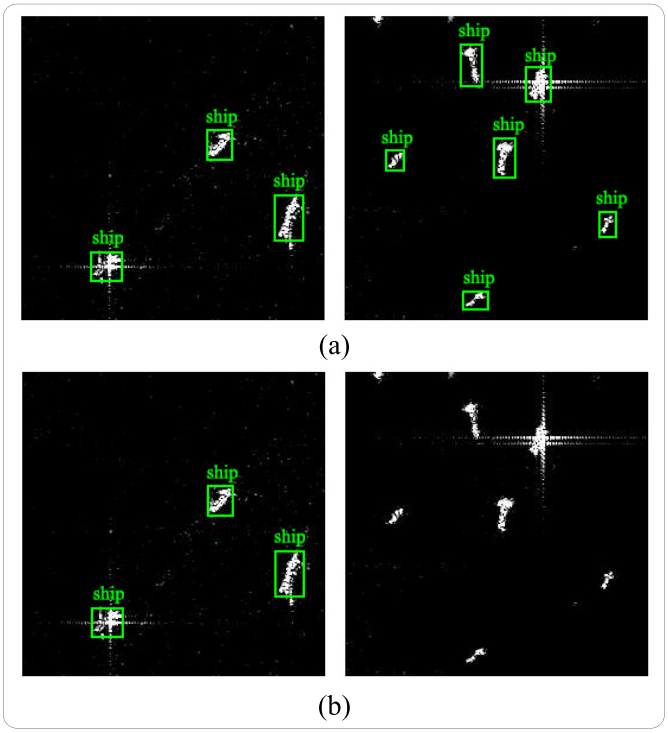


Fig. 1. Difference between (a) supervised learning and (b) semisupervised learning. Supervised learning uses labeled data for training. Semisupervised learning uses labeled data and unlabeled data to jointly train the networks.

and two-stage algorithms. The one-stage algorithms directly regress and classify the object boxes, such as you only look once (YOLO) [16] and the single-shot multibox detector (SSD) [17]. The two-stage algorithms produce region proposals and then calculate the location and classification of the final results according to the region proposals. Region CNN (R-CNN) [18] series algorithms are classic two-stage object detection algorithms.

Training a neural network requires a large amount of data, and the data annotation of object detection needs a lot of manpower and material resources [19]. The imaging principle of SAR requires researchers who annotate data to have the corresponding academic background and experience. It is difficult for inexperienced researchers to accurately mark the correct target in SAR images. SAR systems around the world continue to produce a large number of SAR images every day. Thus, existing labeled SAR ship detection data are rare, while the number of unlabeled SAR images is huge. Compared with supervised learning, semisupervised learning can use both labeled data and unlabeled data at the same time. Fig. 1 shows the difference between supervised learning and semisupervised learning. In supervised learning, all training data are labeled. However, semisupervised learning can use both labeled data and unlabeled data to train the networks. The labeling of SAR images needs experienced experts, which leads to the scarcity of labeled SAR ship data. Therefore, semisupervised learning has more application value than supervised learning in the case of less labeled data. This work focused on semisupervised object detection to propose a semisupervised SAR ship detection network named the semisupervised consistency learning adversarial network (SCLANet).

The two-stage algorithm, which is similar to Faster R-CNN [20], usually contains a region proposal network (RPN). An RPN network can generate the region proposal of an object. According to the region proposals, the features of the object can be extracted in the local area of the image. In our opinion, object detection is the task of extracting interesting objects in local regions. In the complex image scene, the object of interest only occupies a small area; thus, the features of this area are more important than the global background features. The proposed SCLANet can use the region proposals to extract the object features of the local area and judge whether the extracted features are from labeled data or unlabeled data through adversarial learning. The effectiveness of semisupervised learning is usually reflected in the use of clever pretext tasks to train unlabeled data. Consistency learning is proposed, which consists of noise robustness consistency learning and output encoding consistency learning. These two types of consistency learning can serve as pretext tasks. In noise robustness consistency learning, we add noise to the network to make the trained network more robust. Maintaining consistency between the noisy results and the original results can train the unlabeled data. In output encoding consistency learning, an output representation strategy is introduced. The output results are mapped as potential embedding, which plays an auxiliary role in the training process. This potential embedding and a layer before the output layer form an effect similar to an autoencoder. The unlabeled data can be used to train the network according to the errors between these two embeddings.

The contributions of this work are summarized as follows.

1) A new semisupervised SAR ship detection network was proposed.

2) An adversarial network framework for SAR image was designed to make the local feature of ships from an unlabeled image closer to that from a labeled image.

3) A pretext task called noise robustness consistency learning was designed for semisupervised training of SAR images. Noise is added to the SCLANet to learn the consistency between noisy output and noiseless output. Noise robustness consistence learning can not only use unlabeled data to train the SCLANet but can also enhance the robustness of the SCLANet.

4) Another pretext task called output encoding consistency learning was designed. In output encoding consistency learning, the predicted result is transformed into a picture as big as the original image. An encoder maps the picture to an embedding. One layer of the neural network in R-CNN is as another embedding. Reducing the errors between two embeddings can train unlabeled data.

The rest of the article is arranged as follows. The related work is introduced in Section II. Section III shows the framework of the whole SCLANet and describes each module in detail. The experimental results and analysis are presented in Section IV. Finally, Section V summarizes the conclusions.

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II. Related Work

A. Traditional Ship Detection Algorithms

SAR ship detection is a research direction of interest in the field of remote sensing. Many researchers have developed a large number of algorithms. Traditional algorithms can be divided into two categories: one is based on statistical models; the other is based on various features, such as geometric features, contour features, and polarization features. According to different scenes and resolutions of SAR images, researchers have proposed various CFAR algorithms based on statistical models and derivatives of this kind of algorithm. di Bisceglie and Galdi [21] proposed a CFAR model based on the Weibull distribution, which makes it easier to scan targets in high-resolution images through sliding windows. Huang *et al.* [22] fused multiscale features to improve the saliency of local regions and then determined the appropriate threshold to detect ships through a CFAR framework. Ai *et al.* [23] proposed a CFAR detection algorithm that used the 2-D joint log-normal distribution between adjacent pixels to the model. Souyris *et al.* [24] evaluated the polarization and phase of target scatterers in SAR images. They also defined a Hermite product to increase the contrast between the target and the background. Ouchi *et al.* [25] extracted cross correlation values from moving small windows in multilook SAR images to produce a coherent image. Ships can be extracted by using the appropriate threshold because the coherence of ships is higher than that of the sea surface. It is also a common research idea to use object features for detection. Duan *et al.* [26] regarded the ship as an ellipsoid and extracted the three-dimensional geometric features of the ellipsoid. The feasibility and effectiveness of the ellipsoid approximation method were verified by experiments on simulated images. An active contour model that can detect ships efficiently combined the region feature and the edge feature [27]. Fan *et al.* [28] designed a non-Gaussian K-Wishart classifier to divide PolSAR data into different clusters. Ship identification can be realized by combining different clusters with polarization span parameters. Wang and Liu [3] presented a ship detector based on the superpixel-level scattering mechanism distribution features. Gambardella *et al.* [29] considered the ship as a dominant scatterer with a strong coherent backscattering signal and proposed a physical filtering method to detect the ship. Although these traditional algorithms have good detection efficiency, their detection accuracy still needs to be improved in SAR images with complex scenes.

B. Deep Learning Object Detection Algorithms

Deep learning algorithms have greatly changed the field of object detection. The object detection algorithm is usually composed of a backbone, which can extract features, and a head, which can locate and classify the object accurately. Visual geometry group (VGG) [30], ResNet [31], GoogleNet [32], and DenseNet [33] are classic backbone networks. According to the different heads, these detection algorithms are divided into one-stage algorithms and two-stage algorithms. The most classic one-stage algorithms are YOLO [16] and SSD [17]. The first work of the YOLO series algorithm was created by Redmon *et al.* [16] YOLOv2 [34], YOLOv3 [35], and YOLOv4 [36] are constantly reforming this series of algorithms, which also improves the detection accuracy. SSD and its improved algorithms, such as M2Det [37] and RefineDet [38], are also very good one-stage algorithms. Compared with the one-stage algorithm, the two-stage algorithm has more region proposal processes. Although the one-stage algorithm is faster, the two-stage algorithm usually has higher accuracy. Girshick *et al.* proposed several two-stage algorithms, such as R-CNN [18], Fast R-CNN [39], and Faster R-CNN [20]. Whether it is a one-stage algorithm or a two-stage algorithm, these algorithms are basically supervised learning algorithms that cannot use unlabeled data for training. Carion *et al.* [40] used transformers for object detection tasks. Transformers also yielded good results in replacing CNN, which opened up a new idea for researchers.

C. Semisupervised Object Detection

Compared with supervised learning algorithms, semisupervised learning algorithms can use both labeled data and unlabeled data for training. Several studies have applied semisupervised learning to object detection. For example, Gao *et al.* [41] identified three types of noise and presented a novel framework for handling these noises that can improve detection accuracy. Tang *et al.* [42] conducted a detailed study of the visual and semantic similarity of object categories and designed a high-performance object detector based on the combination of visual and semantic similarity. A positioning loss and a noise loss are combined as a pretext task to train the neural network [19]. Zhang *et al.* [43] used adversarial learning to train two classifiers to find complementary object regions. Jeong *et al.* [44] designed two consistency-based loss functions for classification and localization, which can be performed on unlabeled data for semisupervised learning.

D. SAR Ship Detection With CNN

Many deep learning algorithms have been used in SAR ship detection. Zhang *et al.* [45] proposed a lightweight SSD ship detector that includes a novel bidirectional feature fusion module. Lin *et al.* [46] improved Faster R-CNN by using the squeeze excitation mechanism and shared feature maps. A novel SAR ship detection network was designed, which includes a backbone network composed of multiple deny blocks and a feature reuse module [47]. Hou *et al.* [48] proposed a detector that consists of an SSD, a spatial template-matching module, and RefinedNet, and validated this algorithm on multiple datasets. According to the characteristics of the close arrangement of ships, Chen *et al.* [49] used multiscale information and the loss function of a rotated box to train a network. A multiscale ship detection method based on a dense attention pyramid network was proposed [50]. Cui *et al.* [51] proposed an anchor-free method that was added a new spatial shuffle-group enhance attention module to CenterNet to suppress noises. Although there are many deep learning algorithms for SAR ship detection, most of them are supervised learning. These algorithms cannot use unlabeled data for training. Therefore, this work proposed a novel semisupervised SAR ship detection network.

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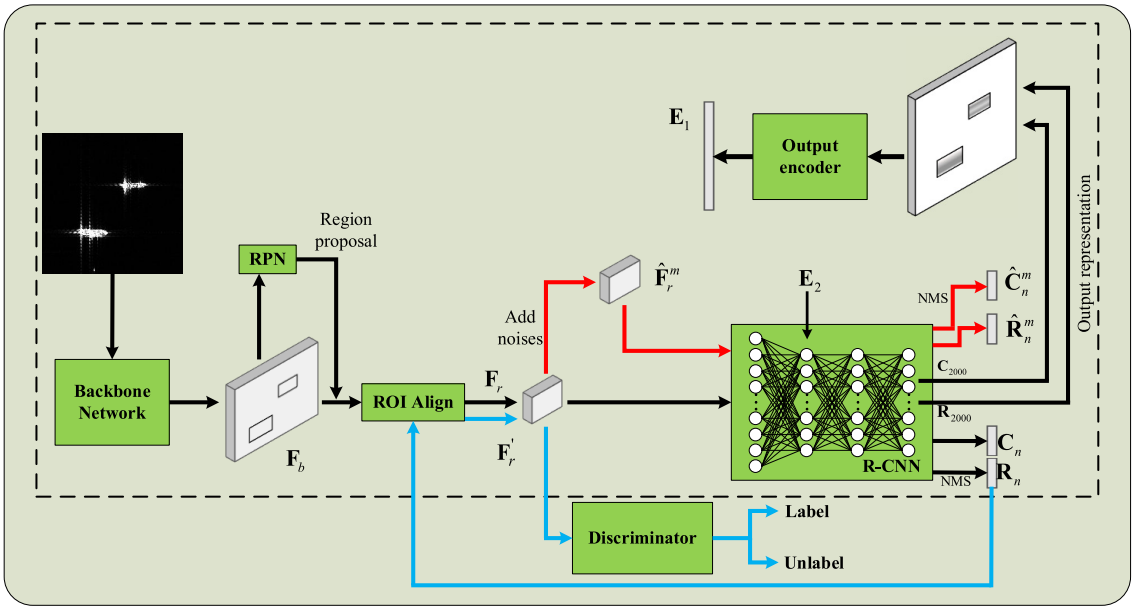


Fig. 2. Architecture of the SCLANet. The SCLANet is divided into a detector and a discriminator. The blue arrows indicate adversarial learning. The red arrows indicate noise robustness consistency learning. The output representation can map the final result to a picture.

III. Method

A. Architecture

The SCLANet, based on Faster R-CNN, is a two-stage algorithm designed for semisupervised SAR ship detection. Fig. 2 shows the architecture of the SCLANet. The improvement of the SCLANet includes the introduction of adversarial learning and consistency learning. In our opinion, the key point of ship detection in a complex scene is the local area around the ship. If the similar local features can be extracted from the labeled data and the unlabeled data, ship detection can be well conducted in the unlabeled image. The same is true of the purpose of introducing adversarial learning into the SCLANet. In Faster R-CNN, RPN can generate a region proposal about the object. The next region of interest alignment (ROIAlign) layer can map local features of different sizes to the same size. He *et al.* [52] proposed the ROIAlign layer and applied it to Faster R-CNN and Mask R-CNN. The local area feature is defined as $\mathbf{F}_r$. After R-CNN and nonmaximum suppression (NMS) processing, the classification results $\mathbf{C}_n$ and location regions $\mathbf{R}_n$ are obtained. Also, n means that there are n bounding boxes. In Fig. 2, the blue arrows indicate adversarial learning. RPN produces a large number of region proposals, which contains a lot of background. Compared with these region proposals, $\mathbf{R}_n$ can frame more accurate local regions of the ship. $\mathbf{R}_n$ is fed back to the ROIAlign layer to generate the feature $\mathbf{F}_r'$. The sizes of $\mathbf{F}_r$ and $\mathbf{F}_r'$ are the same. $\mathbf{F}_r$, used in the training process of the detector, is the feature generated by backbone network and region proposals, while $\mathbf{F}_r'$ is fed to the discriminator, which is in adversarial learning. $\mathbf{F}_r'$ is the feature generated by $\mathbf{C}_n$, $\mathbf{R}_n$, and backbone network. In adversarial learning, $\mathbf{F}_r'$ is fed into the discriminator to determine whether this feature is generated from labeled data or unlabeled data. We hope that the local features ($\mathbf{F}_r'$) extracted by the detector can deceive the discriminator. In the process of training, the discriminator and detector are trained alternately. Consistency learning is divided into two parts: noise robustness consistency learning and output encoding consistency learning. To improve the noise resistance of the SCLANet, the noise robustness consistency learning module is designed. In Fig. 2, the red arrows represent noise robustness consistency learning. $\hat{\mathbf{F}}_r^m$ is obtained by adding noise to feature $\mathbf{F}_r$, and the results of classification $\hat{\mathbf{C}}_n^m$ and location region $\hat{\mathbf{R}}_n^m$ are obtained by sharing R-CNN. Subscript m means adding noise m times. $\mathbf{F}_r$ is fed to the R-CNN to obtain the real outputs $\mathbf{C}_n$ and $\mathbf{R}_n$. A noise consistency loss function is proposed by combining the noisy output with the noiseless output. Noise robustness consistency learning can be a pretext task for semisupervised learning. Another consistency learning module is the output encoding consistency learning. Outputs are represented as pictures that can be input into the neural network. An encoder maps the representation picture to an embedding $\mathbf{E}_1$. We think that one layer of the neural network in the R-CNN can be another embedding $\mathbf{E}_2$. Reducing the error between $\mathbf{E}_1$ and $\mathbf{E}_2$ can achieve an effect similar to that of an autoencoder, which can also be a pretext task for semisupervised learning.

B. Adversarial Learning

The whole framework of SCLANet adopts the way of adversarial learning, which aims to make the detector of SCLANet extract similar features from unlabeled data and labeled data. Feature $\mathbf{F}_b$, extracted from the backbone network, is fed to the RPN. The RPN can then generate a large number of region proposals. In the original Faster R-CNN, the number of region proposals in the training process is about 2000, and the number of region proposals in the inference process is about 1000. According to all region proposals, the ROIAlign layer can map local features of different sizes to

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features of the same size. The size of feature $\mathbf{F}_r$, generated by the ROIAlign, is $2000 \times 256 \times 7 \times 7$ or $1000 \times 256 \times 7 \times 7$. The region proposals generated by RPN are inaccurate and contain a large number of background images. $\mathbf{F}_r$ is the intermediate feature of the detector. This feature is not suitable for adversarial learning. We hope that the results of the detector can deceive the discriminator. Therefore, the features $\mathbf{F}'_r$ that are fed into the discriminator are generated by the result of the detector. The SCLANet is an improvement based on Faster R-CNN that can conduct adversarial learning. The SCLANet follows a part of the classic Faster R-CNN, including a backbone network, an RPN, and an ROIAlign. In the training phase of the SCLANet, the RPN of the SCLANet is extracted to 2000 region proposals. After R-CNN, the results, $\mathbf{C}_{2000}$ and $\mathbf{R}_{2000}$, of the 2000 boxes are output. Then, these boxes only retain the boxes with ship confidence higher than 0.7. The final results, $\mathbf{C}_n$ and $\mathbf{R}_n$, are obtained after the NMS process. $\mathbf{R}_n$ represents only a few of the most accurate bounding boxes. According to the final result, $\mathbf{R}_n$, the ROIAlign layer can extract more accurate local features, $\mathbf{F}'_r$, of size $n \times 256 \times 7 \times 7$. The SCLANet, designed for semisupervised learning, is trained by both unlabeled data and labeled data. We expect that the feature $\mathbf{F}'_r$ extracted from unlabeled data can cheat the discriminator, that is, the closer the features that are extracted from unlabeled data and labeled data, the better. SAR ship detection has only one classification. $\mathbf{F}'_r$ was either the feature of ship or the feature of background. The discriminator discriminated whether $\mathbf{F}'_r$ came from labeled data or unlabeled data, and it also discriminated whether $\mathbf{F}'_r$ was ship feature or background feature. When the discriminator cannot distinguish the authenticity of $\mathbf{F}'_r$, it means that $\mathbf{F}'_r$ extracted from unlabeled data is an accurate feature about the ship. Fig. 3 shows the structure of the discriminator. The discriminator consists of four convolution layers and four full-connection layers. The loss to train the discriminator is

$$L_{\text{ad}} = \sum_{\mathbf{I}\in L} 1 - \text{Dis}\left(\mathbf{F}'_r\right) + \sum_{\mathbf{I}\in U} \text{Dis}\left(\mathbf{F}'_r\right), \mathbf{F}'_r = \text{Bac}(\mathbf{I}) \quad (1)$$

where $\mathbf{I}$ is the original image, L is a set of labeled data, U is a set of unlabeled data, Dis() is the discriminator, Bac() is the composition of the backbone network, and $\mathbf{C}_n$ and $\mathbf{R}_n$ are the ROIAlign layers. For the discriminator, the ground truth of the labeled data is 1, and the ground truth of the unlabeled data is 0.

C. Noise Robustness Consistency Learning

To make the SCLANet more robust, noise robustness consistency learning is used to train the network. Noise robustness consistency learning can also be used as a pretext task for unlabeled data. In Fig. 2, noise robustness consistency learning is represented as red arrows. DropBlock [53] is a way to add noise. We use DropBlock to add m times noise to feature $\mathbf{F}_r$ to generate $\hat{\mathbf{F}}_r^m$. These features ($\mathbf{F}_r, \hat{\mathbf{F}}_r^m$) are fed to the shared network R-CNN to produce the final results, including original outputs ($\mathbf{C}_n$, $\mathbf{R}_n$) and noisy outputs ($\hat{\mathbf{C}}_n^m$, $\hat{\mathbf{R}}_n^m$). The consistency between the original results and the noise results is expected to be ensured. The noisy output should be as close as possible to the original output. The loss can be expressed as follows:

$$L_{\text{noise}} = \frac{1}{N}\sum_{n=1}^{N}\frac{1}{M}\sum_{m=1}^{M}\left\|\mathbf{C}_n - \hat{\mathbf{C}}_n^m\right\|_2 + \left\|\mathbf{R}_n - \hat{\mathbf{R}}_n^m\right\|_1. \quad (2)$$

We use the *L2* norm to optimize the classification outputs and the *L1* norm to optimize the bounding box outputs. Noise robustness consistency learning can not only improve the robustness of the SCLANet but can also be used as a pretext task of semisupervised learning to train unlabeled data.

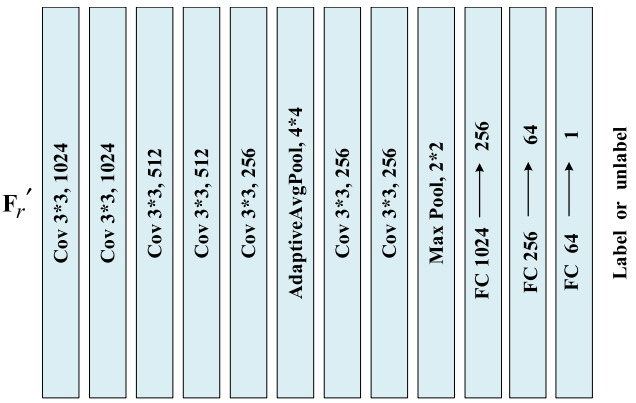


Fig. 3. Structure of the discriminator.

D. Output Encoding Consistency Learning

Output encoding consistency learning is another designed pretext task. The core is to define the function $A(\cdot)$

$$\begin{aligned} &A(f(\mathbf{I}, \mathbf{W}_f), \mathbf{Y}) = L_{\text{dis}}(\mathbf{E}_2, \mathbf{E}_1) \\ &\mathbf{E}_2 = f(\mathbf{I}, \mathbf{W}_f), \mathbf{E}_1 = \text{Enc}(\mathbf{Y}, \mathbf{W}_e) \end{aligned} \quad (3)$$

where $\mathbf{I}$ is the original image, $f(\cdot)$ is the network in front of embedding $\mathbf{E}_2$, and $\mathbf{W}_f$ correspond to the corresponding weight. Enc$(\cdot)$ is the output encoder, and $L_{\text{dis}}(\cdot)$ is the distance measurement function. $\mathbf{W}_e$ is the output encoder and output representation. For convenience, we use $\mathbf{Y}$ for outputs ($\mathbf{C}_n$ and $\mathbf{R}_n$). In addition, the process of embedding $\mathbf{E}_2$ forming $\mathbf{Y}$ is in the R-CNN and can be expressed as follows:

$$\mathbf{Y} = r(\mathbf{E}_2, \mathbf{W}_r) \quad (4)$$

where $\mathbf{W}_r$ is the weight of R-CNN. Ideally, the loss of $A(\cdot)$ should be close to zero. The part of the SCLANet between these two embeddings can be regarded as an autocoder. $\mathbf{E}_1$ and $\mathbf{E}_2$ can be used as the intermediate representations of the output.

Equation (3) cannot be calculated directly. Thus, we first assume that we have the optimal model and the corresponding optimal weights $\mathbf{W}_f^*$ and $\mathbf{W}_r^*$. The inverse of $r(\cdot)$

$$\mathbf{E}_2 = \mathbf{E}_1 = r^{-1}\left(\mathbf{Y}, \mathbf{W}_r^*\right) \quad (5)$$

is the reverse process of the output result $\mathbf{Y}$ to intermediate embedding. We introduce a network, Enc$(\cdot)$, the weights of

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which can be learned to simulate $r^{-1}(\cdot)$. The optimization process is given as follows:

$$\mathbf{W}_e^* = \arg\min_{\mathbf{W}_e} L\left(r\left(\text{Enc}(\mathbf{Y}, \mathbf{W}_e), \mathbf{W}_r^*\right), \mathbf{Y}\right). \tag{6}$$

$L(\cdot)$ is the detection loss, which will be introduced in the next section. It can be seen that the output encoder Enc$(\cdot)$ maps the output results to the potential feature space, and $r(\cdot)$ maps the feature to the output results. We rewrite $A(\cdot)$ as follows:

$$A(f(\mathbf{I}, \mathbf{W}_f), \mathbf{Y}) = L_{\text{dis}}(f(\mathbf{I}, \mathbf{W}_f), \text{Enc}(\mathbf{Y}, \mathbf{W}_e)). \tag{7}$$

The above inference is based on the assumption of having optimal weight. In the actual training model, we use two tricks to overcome this difficulty. One is to use the pretraining model, which is trained with labeled data. Another trick is to use (6) and (7) alternately as the loss function of output encoding consistency learning. Here, we rearrange the loss function of output encoding consistency learning as follows:

$$L_{\text{output}} = \begin{cases} L(r(\text{Enc}(\mathbf{Y}, \mathbf{W}_e), \mathbf{W}_r^*), \mathbf{Y}), & \text{iter} = 1, 3, 5, \ldots \\ L_{\text{dis}}(f(\mathbf{I}, \mathbf{W}_f), \text{Enc}(\mathbf{Y}, \mathbf{W}_e)), & \text{iter} = 2, 4, 6, \ldots \end{cases} \tag{8}$$

where $L_{\text{dis}}(\cdot)$ is the $L2$ distance.

The final outputs include the location $\mathbf{R}_{2000}$ and classification $\mathbf{C}_{2000}$ of 2000 boxes. These outputs can generate a picture that is used as the input of network Enc$(\cdot)$ through output representation. Specifically, a zero matrix with the same size as the original image is generated first. According to the coordinates of each bounding box, $\mathbf{R}_i$, the confidence value, $\mathbf{C}_i$, or its opposite number is filled in the corresponding region of the zero matrix. $\mathbf{R}_{2000}$ and $\mathbf{C}_{2000}$ contain not only the results of the background boxes but also the results of the ship boxes. Thus, the output representation follows three principles: 1) the result of recognition as the background is filled first, and the result of recognition as the ship is filled later; 2) the order of confidence, from small to large, is filled; and 3) the area corresponding to the background box is filled with the opposite number of the confidence, and the area corresponding to the ship box is filled with the confidence value. Fig. 4(a) shows the schematic of the output representation. The picture generated by the output representation is a matrix with a maximum value of 1 and a minimum value of -1. Values between -1 and 0 represent the background, and values between 0 and 1 represent the ship area. This picture is a bit similar to the average SAR image and highlights the ship area. This makes it easier to learn the intermediate representation of the output. The generated picture is fed into the output encoder, Enc$(\cdot)$, to generate embedding $\mathbf{E}_1$. The output encoder consists of four convolution layers, two max-pooling layers, and a full-connection layer, as shown in Fig. 4(b).

In the original Faster R-CNN, the R-CNN module only contains two full-connection layers. In the SCLANet, we add two full-connection layers to the R-CNN module. Embedded $\mathbf{E}_2$ is marked in Fig. 2. When unlabeled data are fed to the SCLANet, embeddings $\mathbf{E}_1$ and $\mathbf{E}_2$ are obtained. Output encoding consistency learning is also a pretext task for semi-supervised SAR ship detection.

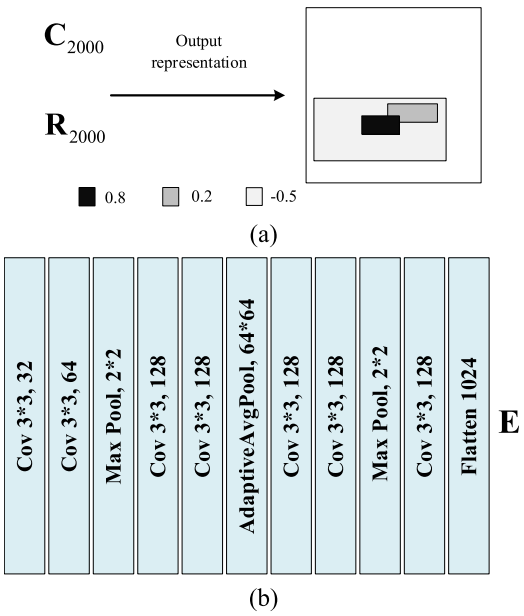


Fig. 4. (a) Schematic about output representation. For example, the confidence of the black region is 0.8, that of the gray area is 0.2, and that of the light gray area is -0.5. (b) Structure of the encoder.

E. Training

In this section, the process of training SCLANet is described in detail. Unlike supervised learning, semisupervised learning needs to utilize a large amount of unlabeled data and labeled data to train the network. The detection loss of Faster R-CNN can be used to train the SCLANet with labeled data

$$L = L_c + L_r. \tag{9}$$

L_c is the classification loss

$$L_c = -\log c \tag{10}$$

where c is the probability value of the true class and L_r is the bounding box regression loss

$$L_r = \sum_{i \in \{x,y,w,h\}} \text{smooth}_{L_1}(t_i - v_i)$$
$$\text{smooth}_{L_1}(x) = \begin{cases} 0.5x^2, & \text{if } |x| < 1 \\ |x| - 0.5, & \text{other} \end{cases} \tag{11}$$

where $v = (v_x, v_y, v_w, v_h)$ are regression targets and $t = (t_x, t_y, t_w, t_h)$ are predicted results.

There are three steps in the training of SCLANet. The first step is to train the detector of the SCLANet with labeled data. The loss function is

$$L_1 = L. \tag{12}$$

Although consistency learning can be applied to training with labeled data, we only use consistency learning to train the SCLANet with unlabeled data. The second step is to train the detector of SCLANet with unlabeled data. The loss function is

$$L_2 = \alpha_1 L_{\text{noise}} + \alpha_2 L_{\text{output}} - \alpha_3 L_{\text{ad}}. \tag{13}$$

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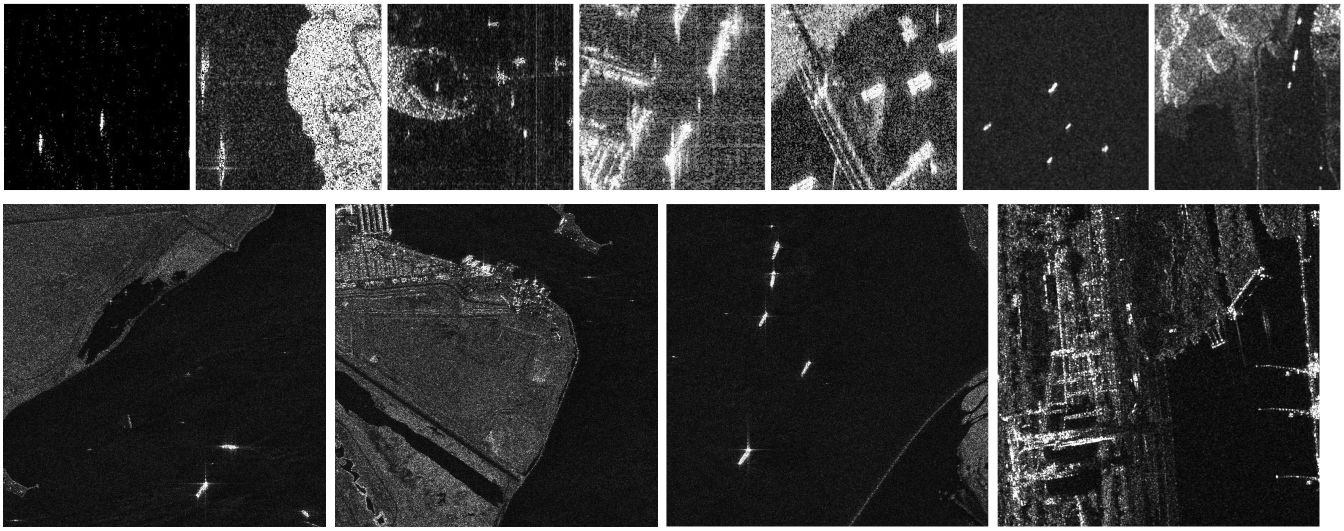


Fig. 5. Examples of two datasets. The first row is some images of SAR-Ship-Dataset. The second row is some images of HRSID.

The last step is to train the discriminator of the SCLANet. Both unlabeled data and labeled data train discriminators. The loss function is

$$L_3 = L_{\text{ad}}. \tag{14}$$

The detector and the discriminator are trained alternately. During the inference time, the SCLANet only retains the same structure as the Faster R-CNN, including the backbone network, the RPN, the ROIAlign, and the R-CNN. In addition, we used labeled data to train the SCLANet for five epochs to obtain a pretraining model. The pretraining model can effectively reduce the loss of nonconvergence caused by pretext tasks.

IV. Experiments

A. Experimental Setup

SCLANet and comparison algorithms were performed on two datasets. Wang *et al.* [54] constructed an SAR ship detection dataset (SAR-Ship-Dataset) that contained 43 819 SAR image chips with a length and a width of 256. These image chips were from 102 Gaofen-3 SAR images and 108 Sentinel-1 SAR images. The whole dataset contained about 30 000 ship objects. These images contained complex scenes: ports, islands, and coastal waters. The distribution of ships was also in various forms, some of which were very scattered, while some were very concentrated. This dataset was annotated using the standard visual object classes (VOC) format. In Fig. 5, the top row shows some images in SAR-Ship-Dataset. The second dataset is a high-resolution SAR images dataset (HRSID) [55]. HRSID contained 5604 SAR images. These images were with a length and a width of 800 and a resolution ranging from 1 to 5 m. The SAR images in HRSID were from Sentinel-1B, TerraSAR-X, and TanDEMX satellites. They were annotated in the format of Microsoft common objects in context (COCO). The bottom row of Fig. 5 is the example images.

TABLE I
COCO Metrics

Metric	Meaning
AP	AP at IOU = 0.50: 0.05: 0.95
AP_{50}	AP at IOU = 0.50
AP_{75}	AP at IOU = 0.75
AP_S	AP for small objects: area < 32^2 (IOU = 0.50: 0.05: 0.95)
AP_M	AP for medium objects: 32^2 < area < 96^2 (IOU = 0.50: 0.05: 0.95)
AP_L	AP for large objects: area > 96^2 (IOU = 0.50: 0.05: 0.95)

The COCO index is a classic evaluation index in the field of object detection. The COCO metrics were used to calculate the intersection over union (IOU) between the predicted result and the ground truth. According to the different IOUs, the mean average precision (AP) of results was calculated. Table I lists the COCO metrics.

The proposed SCLANet was compared with other algorithms: Faster R-CNN [20], YOLOv4 [36], CSD-SSD [44], Unbiased-teacher [56], STAC [57], and UFO2 [58]. Yolov4 and Faster R-CNN are supervised learning algorithms, which can only use labeled data for training. The other four comparison algorithms are semisupervised learning algorithms, which can use labeled data and unlabeled data for training. The experiments in this study were performed on an NVIDIA DGX-station with TeslaV100 GPU, Python 3.6, and Intel Xeon E5-2698 CPU.

B. Experiments on SAR-Ship-Dataset

In the SAR-Ship-Dataset experiments, we randomly divided all the images into 30% as the test set, 30% as the labeled training set, and 40% as the unlabeled training set. The batch size was 2. The detector and the discriminator were optimized using random gradient descent (RGD). The learning rate of the detector was $1e-4$. The learning rate of the discriminator was $5e-5$.

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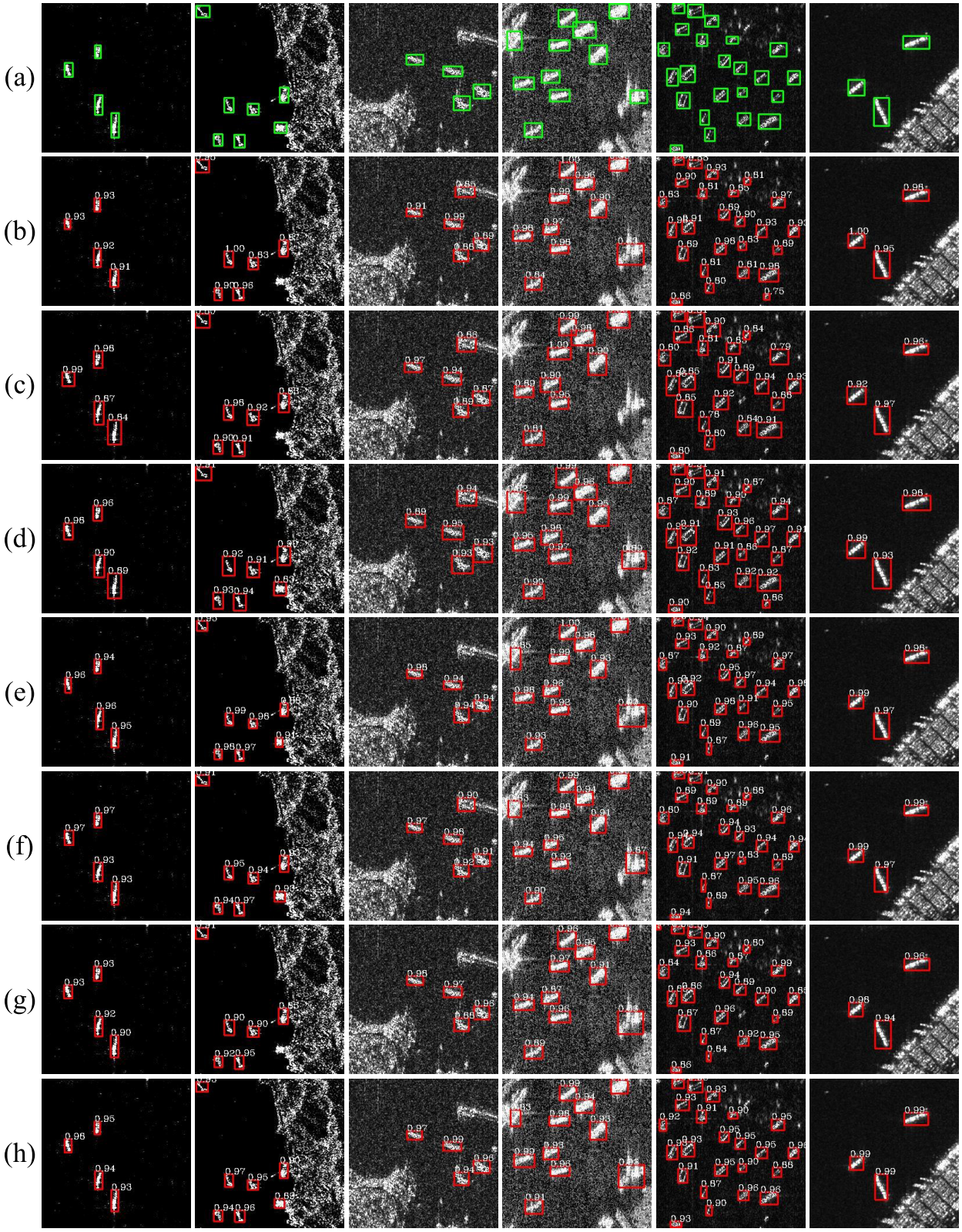


Fig. 6. Experimental results on SAR-Ship-Dataset. (a) First row is the ground truth. (b) Second row shows the results of Faster R-CNN. (c) Third row shows the results of YOLOv4. (d) Fourth row shows the results of CSD-SSD. (e) Fifth row shows the results of Unbiased-teacher. (f) Sixth row shows the results of STAC. (g) Seventh row shows the results of UFO2. (h) Last row shows the results obtained by the SCLANet.

Fig. 6 shows the results of all the algorithms on the SAR-Ship-Dataset. Each row is the result of an algorithm. In Fig. 6, the green boxes are the ground truth, and the red boxes are the prediction results of the algorithms. The score above each red box indicates the confidence of the detected ship. In training Faster R-CNN, only 30% of data were used.

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The inference result of Faster R-CNN was not accurate. Some small ships and berthing ships were easily missed. Many types of sea and shore clutter were identified as ships. For example, there were missing ships landing on the shore in the second and fourth columns of Fig. 6(b); many sea clutters were mistakenly detected as ships in the fifth column of Fig. 6(b). YOLOv4 is a supervised learning algorithm, such as Faster R-CNN. Because less labeled data can be used, the accuracy of YOLOv4 was also relatively low. Many false checks and omissions appeared in the inference results of YOLOv4. Comparing Fig. 6(b) and (c), the confidence of YOLOv4 to identify ships was lower than that of Faster R-CNN. The confidence of the two ships was below 0.9 in the first column of Fig. 6(c), while the results of Faster R-CNN were all above 0.9. Overall, the performance of YOLOv4 was poorer than that of Faster R-CNN. CSD-SSD, Unbiased-teacher, STAC, and UFO2 are semisupervised algorithms. They can use both labeled data and unlabeled data in the training process. The performance of these semisupervised algorithms was better than that of supervised algorithms. CSD-SSD is essentially a single-stage detector used as a consistency constraint to optimize ship classification and location. Compared with the previous two supervised learning algorithms, the case of missed detection was less in the results of CSD-SSD. There was a ship missing in the result of Faster R-CNN [the fourth column of Fig. 6(b)], but there was no missing in the result of CSD-SSD [the fourth column of Fig. 6(d)]. Unbiased-teacher is a kind of dual network (a teacher network and a student network) mutual benefit training mode. Both the student network and the teacher network are Faster R-CNNs. Unbiased-teacher performed much better than the original Faster R-CNN. In the results of Unbiased-teacher, the situation of the missed inspection was rare, and a small amount of clutter was identified as ships. Moreover, the bounding boxes predicted by Unbiased-teacher fit the ship more tightly than those predicted by CSD-SSD. In the first two columns of Fig. 6(d) and (e), the prediction boxes of Unbiased-teacher were tighter surrounding ships, and the prediction boxes of CSD-SSD were larger. In other words, Unbiased-teacher's positioning boxes were more accurate. STAC uses a data enhancement strategy and high-confidence pseudolabels for semisupervised training. STAC is also implemented on the basis of Faster R-CNN. The performance of STAC was similar to that of Unbiased-teacher, and it would be interfered with by some sea clutter to result in false detection. UFO2 is a unified framework of omnisupervised object detection. It can do semisupervised learning, weakly supervised learning, and so on. Although the performance of UFO2 was better than that of Faster R-CNN and YOLOv4, false detection and missed detection still appeared in the results of UFO2. Compared with other algorithms, the proposed SCLANet was more robust and less affected by sea clutter. False detection and missed detection were rare in the results of the SCLANet. Moreover, the confidence of the ship detected by the SCLANet was higher than that of other algorithms. In general, the performance of the SCLANet was better than that of other algorithms.

TABLE II
AP OF DIFFERENT METHODS ON SAR-SHIP-DATASET

	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
Faster R-CNN	0.4204	0.7902	0.3994	0.3704	0.4840	0.3144
YOLOv4	0.3827	0.6989	0.3028	0.3363	0.4487	0.2123
CSD-SSD	0.5007	0.8060	0.4881	0.4578	0.5757	0.4319
Unbiased-teacher	0.5217	0.8545	0.5127	0.4796	0.5969	0.4633
STAC	0.5208	0.8379	0.4874	0.4611	0.6012	**0.4771**
UFO2	0.4886	0.7966	0.4688	0.4262	0.5578	0.4190
SCLANet	**0.5420**	**0.8825**	**0.5341**	**0.4847**	**0.6209**	0.4226

The quantitative analysis results of all algorithms are presented in Table II. The AP values of semisupervised algorithms were greater than 0.48, while the AP values of supervised algorithms were less than 0.43. In the comparison of the two supervised algorithms, the performance of Faster R-CNN with an AP value of 0.4204 was better than that of YOLOv4. The AP value of UFO2 was 0.4886, which was the lowest in the semisupervised algorithm. The AP of CSD-SSD was 0.0121 higher than that of UFO2. The performance of Unbiased-teacher was similar to that of STAC. The AP value of one was 0.5217, and that of the other was 0.5208. The AP value of the SCLANet was the highest, at 0.5420. In addition to AP, AP_{50} and AP_{75} are also important indexes to evaluate object detection. The AP_{50} of SCLANet was 0.8825, and the AP_{75} of the SCLAnet was 0.5341. The AP_{50} of CSD-SSD, Unbiased-teacher, STAC, and UFO2 was higher than 0.79, at 0.8060, 0.8545, 0.8379, and 0.7966, respectively. Moreover, their AP_{75} values were 0.4881, 0.5127, 0.4874, and 0.4688, respectively. The results of the supervised algorithms were inferior to those of the semisupervised algorithms. The AP_{50} and AP_{75} of the supervised algorithm were less than 0.8 and 0.4, respectively. The AP_S, AP_M, and AP_L were evaluated for objects of different sizes. In this SAR dataset, the number of large objects only accounted for 0.3% of the total data. The AP_L values of CSD-SSD, Unbiased-teacher, and STAC were higher than that of the SCLANet. However, the effect of small object detection and medium object detection was better in the SCLANet. The AP_S value and the AP_M value of SCLANet were 0.4847 and 0.6209, respectively. In general, the performance of the SCLANet was the best. Among the other semisupervised algorithms, Unbiased-teacher and STAC had a better performance. The results of Faster R-CNN and YOLOv4 were significantly inferior to those of the semisupervised algorithms.

To measure the performance of the proposed SCLANet in different scenarios, we divided SAR-Ship-Dataset into inshore images and offshore images. There were 8611 inshore images and 35 208 offshore images. We used the ratio of 3 to 3 to 4 to split the inshore images into a test set, a labeled training set, and an unlabeled training set. The offshore images were also divided in this way. Both for the training of inshore images and training of offshore images, the batch size was 2. The detector and discriminator were optimized using RGD. In the training of inshore images, the learning rate of the detector was $1e-4$, and that of the discriminator was $5e-5$. In the training of offshore images, the learning rate of the detector did not

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TABLE III
AP ABOUT INSHORE IMAGES

	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
Faster R-CNN	0.3672	0.7153	0.3101	0.3025	0.4561	0.3656
YOLOv4	0.2891	0.5893	0.2043	0.2268	0.3795	0.3069
CSD-SSD	0.3859	0.7371	0.3358	0.3198	0.4756	0.3600
Unbiased-teacher	0.4028	0.7552	0.3560	0.3352	0.4932	**0.3768**
STAC	0.3995	0.7508	0.3602	0.3378	0.4874	0.3654
UFO2	0.3794	0.7204	0.3303	0.3150	0.4646	0.3512
SCLANet	**0.4110**	**0.7634**	**0.3712**	**0.3421**	**0.4998**	0.3590

TABLE IV
AP ABOUT OFFSHORE IMAGES

	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
Faster R-CNN	0.6023	0.9270	0.6988	0.5449	0.6666	0.6000
YOLOv4	0.5443	0.8594	0.6325	0.5028	0.5859	0.5832
CSD-SSD	0.6011	0.9178	0.6847	0.5401	0.6710	0.5723
Unbiased-teacher	0.6152	0.9360	0.7096	0.5583	0.6779	**0.6185**
STAC	0.6184	0.9401	0.7113	0.5598	0.6802	0.6053
UFO2	0.5975	0.9122	0.6839	0.5328	0.6566	0.5769
SCLANet	**0.6206**	**0.9529**	**0.7251**	**0.5634**	**0.6874**	0.6089

change, while the learning rate of the discriminator changed to $1e-5$. There was a large contrast between the ships and the sea surface in the offshore image. Ships in offshore images were easier to detect than those in inshore images. This makes it easy for the discriminator to beat the detector and break the Nash equilibrium. Therefore, we used a smaller learning rate to the discriminator when training with offshore images.

Tables III and IV show the quantitative results of inshore images and offshore images, respectively. In Table III, the AP value of all algorithms is not higher than 0.45. The scene of inshore images was complex. The buildings on land produced a lot of clutter that interfered with all algorithms. SCLANet had the highest AP value of 0.4110. The detection performance of SCLANet for large objects was unstable due to the very small amount of large ships and the fact that SCLANet was trained by adversarial learning. However, other indicators of SCLANet were higher than all comparison algorithms. On the whole, SCLANet had the best performance. By comparing AP values, the performance of semisupervised algorithms was obviously better than that of supervised algorithms.

The ships in the offshore images were in great contrast to the sea surface. The sea surfaces were black, and the highlighted area was ships. Ships in offshore images were easier to detect than those in inshore images. It is obvious that the AP and AP_{50} in Table IV were higher than those in Table III. Even all AP and AP_{50} were higher than 0.85 and 0.54, respectively. In Table IV, all algorithms showed good detection results. The AP value of Faster R-CNN was 0.6023, which is higher than that of YOLOv4, CSD-SSD, and UFO2. The AP value of YOLOv4 was the lowest, but it was still higher than 0.5. CSD-SSD and UFO2 were two of the worst semisupervised algorithms. SCLANet had the highest AP value and the best performance.

C. Experiments on HRSID

We also conducted experiments on HRSID. We randomly divided all the images into 30% as the test set, 30% as the labeled training dataset, and 40% as the unlabeled training dataset. On HRSID, the image size was 800×800. SCLANet needs to use images of size 256×256 for training and testing. Therefore, we cut the images to 256×256. We used the step size of 200 for clipping and added padding in the part beyond the original images. The comparison algorithms were also trained with the cropped images. When training SCLANet, the batch size was 2, the learning rate of the detector was $1e-4$, and the learning rate of the discriminator was $5e-5$.

Figs. 7 and 8 show the results of all the algorithms on HRSID. For illustration purposes, each column corresponds to the results of an algorithm in Figs. 7 and 8. Fig. 7(a) shows the ground truth. Fig. 7(b) and (c) correspond to the results of two supervised algorithms (Faster R-CNN and YOLOv4), respectively. It can be seen from the two images in the first row of Fig. 7(b) and (c) that all offshore ships were detected, and the confidence of the boxes was above 0.9. In the results of Faster R-CNN and YOLOv4, a large number of false detection and missing detection occurred in the detection of inshore ships. In the third row of Fig. 7(b), Faster R-CNN detected only three ships, and one of them was detected incorrectly. In the third row of Fig. 7(c), there were a lot of false detections because YOLOv4 was affected by land clutter. In the last row of Fig. 7, Faster R-CNN generated multiple boxes for the detection of large objects, while YOLOv4 missed detection for a large ship. Fig. 7(d) shows the result of the semisupervised algorithm CSD-SSD. The false detection and missed detection of CSD-SSD were less than that of Faster R-CNN and YOLOv4. In the first row of Fig. 7(d), CSD-SSD can accurately detect all offshore ships. As seen in the third row of Fig. 7(d), CSD-SSD detected five ships more accurately than Faster R-CNN and two false detections less than YOLOv4. The performance of CSD-SSD was better than that of supervised algorithms. However, the confidence of the ships identified by CSD-SSD was lower compared to the results shown in Fig. 8. Fig. 8 illustrated the detection results of the semisupervised algorithms. Fig. 8(a) shows the results of Unbiased-teacher, Fig. 8(b) shows the results of STAC, Fig. 8(c) shows the results of UFO2, and Fig. 8(d) shows the results of SCLANet. These four algorithms showed strong detection performance. However, their detection results of ships docked together were poor. In the second row of Fig. 8, there were three ships connected to the dock together in the original image. Unbiased-teacher, UFO2, and SCLANet detected only one of the three ships; STAC missed all three ships. In the last row of Fig. 8, the results of UFO2 and SCLANet for large ships were inaccurate. UFO2 detected only half of the large ship. The result of SCLANet was multiple boxes, while the detection results of Unbiased-teacher and STAC were better for large ships. In addition, these four algorithms were easy to mistakenly detect the shore clutter as a ship. In general, SCLANet had fewer false detections and missed detections than other algorithms.

Table V shows the quantitative analysis results of all algorithms. The AP values of supervised algorithms were lower than 0.5, while the AP values of all semisupervised algorithms were greater than 0.5. YOLOv4 had the lowest metrics compared to the other algorithms. The AP value of YOLOv4

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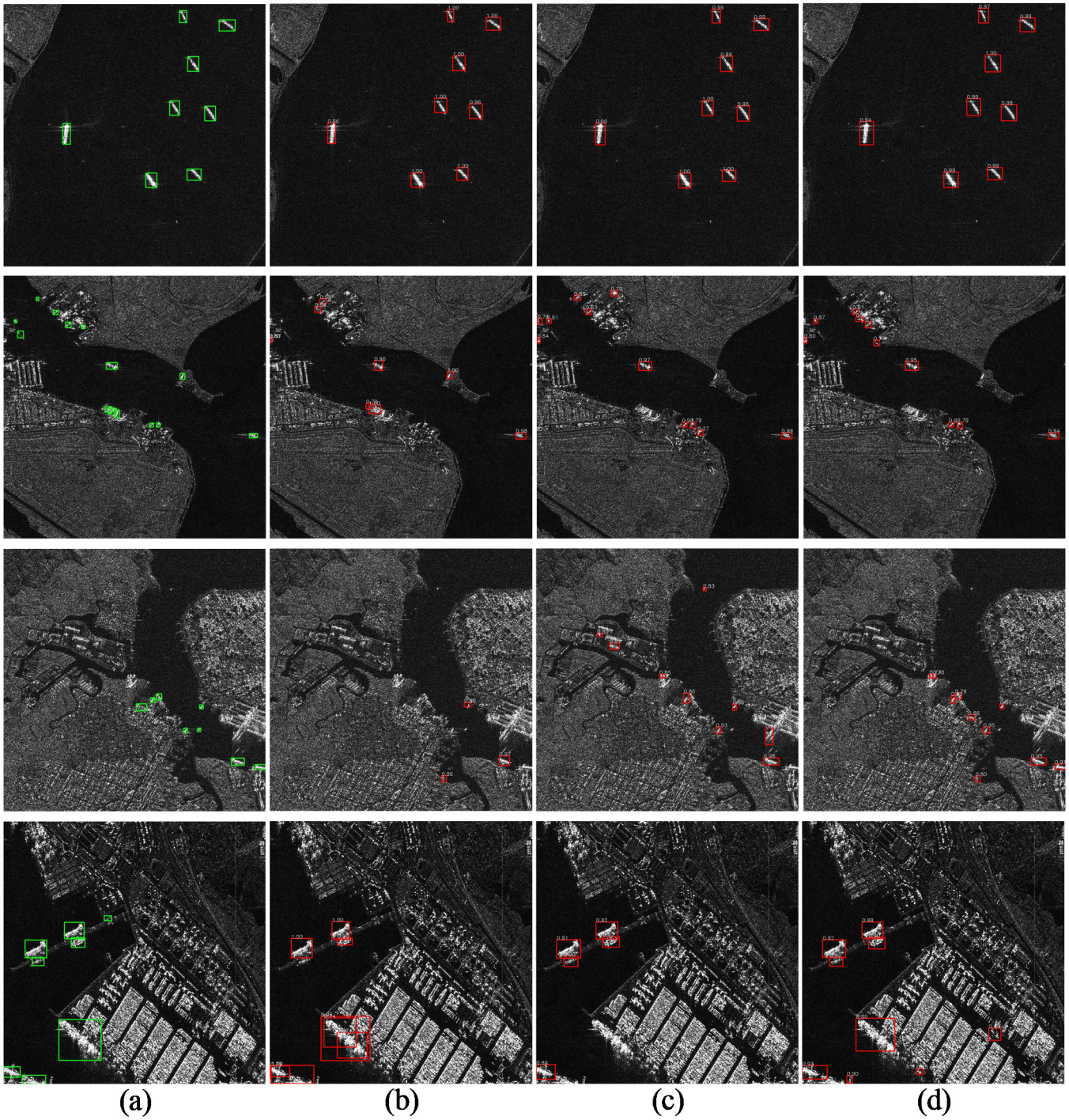


Fig. 7. Experimental results on HRSID. (a) First column is the ground truth. (b) Second column shows the results of Faster R-CNN. (c) Third column shows the results of YOLOv4. (d) Fourth column shows the results of CSD-SSD.

was 0.4543, AP_{50} was 0.7011, and AP_{75} was 0.5023. The AP value of Faster R-CNN was 0.4913, which was higher than YOLOv4. The performance of Faster R-CNN was better than YOLOv4. The detection results of UFO2 and CSD-SSD were worse than that of the other three semisupervised algorithms. However, UFO2 and CSD-SSD outperformed Faster R-CNN and YOLOv4. The AP value of UFO2 was 0.5157, AP_{50} was 0.7963, and AP_{75} was 0.5869. The AP, AP_{50}, and AP_{75} of CSD-SSD were 0.5249, 0.8310, and 0.5994 respectively. The AP value of CSD-SSD was 0.0092 higher than that of UFO2 but 0.0153 lower than that of STAC. Both STAC and Unbiased-teacher had AP values higher than 0.54. This shows that their performance is better than CSD-SSD and UFO2. There was a gap of 0.0066 between the AP value of STAC and that of Unbiased-teacher. STAC was slightly worse than Unbiased-teacher. As can be seen from Table V, each

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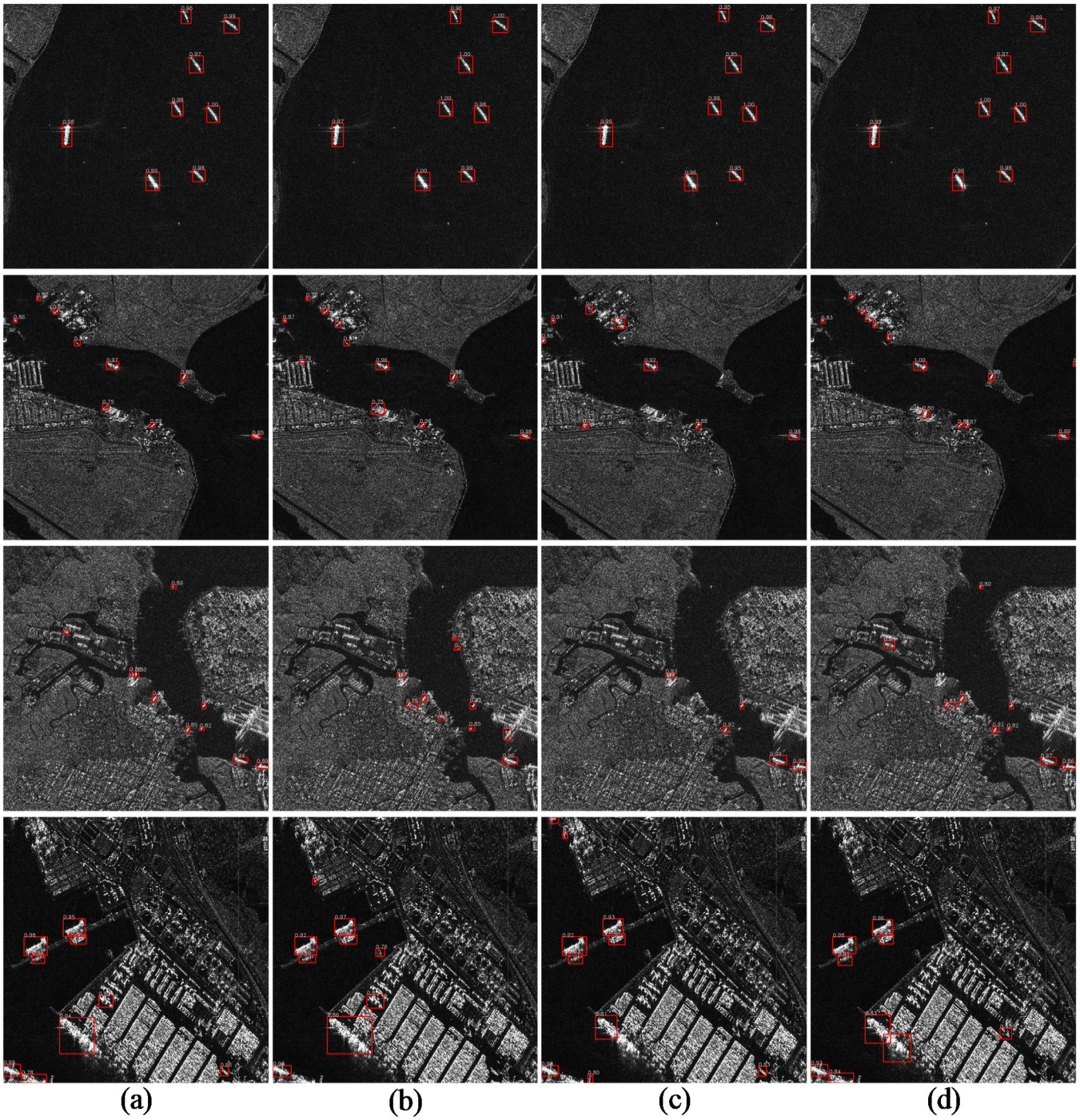


Fig. 8. Experimental results on HRSID. (a) First column shows the results of Unbiased-teacher. (b) Second column shows the results of STAC. (c) Third column shows the results of UFO2. (d) Fourth column shows the results obtained by the SCLANet.

evaluation index of SCLANet was the highest, marked in bold font. The AP value of SCLANet was 0.0116 higher than that of Unbiased-teacher. The AP_{50} of SCLANet was also 0.0189 higher than that of Unbiased-teacher. The AP_{75} of SCLANet was also the highest. This proved that the performance of SCLANet was better than other algorithms.

D. Experiments for Performance Comparison

To study the performance of SCLANet with different amounts of labeled data, we also conducted experiments with different amounts of labeled data. The experiments were done on SAR-Ship-Dataset. We randomly selected 70% of the data as the training set and 30% of data as the test set, and the training data were divided into labeled training data and unlabeled training data. In experiments, the proportion of labeled training data was 5%, 10%, 15%, and 20% of the total data. The rest of the training data were unlabeled. Table VI presents the results of all algorithms with different proportions of labeled data. In this group of experiments, there were less labeled data. The lack of labeled data led to the instability of training.

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TABLE V
AP OF DIFFERENT METHODS ON HRSID

	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
Faster R-CNN	0.4913	0.7569	0.5632	0.3204	0.6917	0.3087
YOLOv4	0.4543	0.7011	0.5023	0.3102	0.6317	0.3010
CSD-SSD	0.5249	0.8310	0.5994	0.3655	0.7236	0.3427
Unbiased-teacher	0.5468	0.8425	0.6121	0.4189	0.7529	0.3768
STAC	0.5402	0.8378	0.6088	0.4026	0.7446	0.3835
UFO2	0.5157	0.7963	0.5869	0.3502	0.7078	0.3622
SCLANet	**0.5584**	**0.8614**	**0.6365**	**0.4328**	**0.7621**	**0.3783**

The use of adversarial learning in SCLANet also led to the unstable training effect. This may cause the ship detection result of SCLANet with high confidence to be worse than other algorithms. In the 5% and 10% cases, the AP_{75} of Unbiased-teacher was higher than that of the SCLANet. However, the evaluation index of SCLANet was the highest in other cases. Overall, the performance of Unbiased-teacher was second only to that of the SCLANet. Also, the semisupervised algorithms were better than the supervised algorithms. With the increase in labeled data, the performance of the supervised learning algorithms was close to that of the semisupervised algorithms. For example, in the case of 5%, the difference between the AP value of the Faster R-CNN and that of the CSD-SSD was 0.0855, and in the case of 20%, the difference between the AP value of the Faster R-CNN and that of the CSD-SSD was 0.062. In addition, the random partition of datasets would affect the supervised learning algorithms with less labeled data. The AP values of Faster R-CNN and YOLOv4 at 5% were higher than those at 10%. The semisupervised algorithms were not affected by random partition, and their robustness was better.

We also counted the average inference time of each algorithm. Table VII shows the statistical results. The inference time was obtained in a TeslaV100 GPU. The batch size at the time of inference was 1. The average inference time for Faster R-CNN was 0.0460616 s. The average inference time for YOLOv4 was 0.0353485 s. The one-stage algorithm YOLOv4 was faster than the two-stage algorithm Faster R-CNN. The average inference time of YOLOv4 was 0.0107131 s less than that of Faster R-CNN. Semisupervised algorithm CSD-SSD was based on SSD. Therefore, CSD-SSD is fast in the inference phase. Its inference time was only 0.0208858 s. The semisupervised algorithms, Unbiased-teacher, STAC, and UFO2, were all based on Faster R-CNN. Although their training frameworks were different, their inference networks were all Faster R-CNN in test time. Therefore, their inference time was the same as that of Faster R-CNN. SCLANet was also based on Faster R-CNN, but with two fully connected layers added to the network structure. The inference time of SCLANet was the longest, which was 0.0462082 s. This showed that the inference speed of SCLANet was similar to that of Faster R-CNN.

E. Ablation Study and Parameter Analysis

We performed ablation experiments to analyze each module of the SCLANet. Both noise robustness consistency learning and output encoding consistency learning can be used as a pretext task for training unlabeled data. In Case 1, the original Faster R-CNN was only trained through noise robustness consistency learning. In Case 2, the original Faster R-CNN was only trained through output encoding consistency learning. In Case 3, the two consistency learning modules were used together. Adding the adversarial learning module on the basis of Case 3 resulted in the complete SCLANet, that is, Case 4. We used 30% of the total data on SAR-Ship-Dataset as labeled training data and 40% as unlabeled training data to on SAR-Ship-Dataset train these four cases. Table VIII presents the results for the four cases. The AP value of Case 1 was the lowest, and that of Case 4 was the highest. The AP value of Case 2 was 0.0234 higher than that of Case 1, which indicates that the training effect of using output encoding consistency learning alone is better than that of using noise robustness consistency learning alone. When two consistency learning modules were used together, the AP value of Case 3 was increased by 0.0406. The gap between the AP value of Case 3 and Case 4 was 0.0185, and the improvement of AP value was caused by adversarial learning. In Table VIII, other indicators from Case 1 to Case 4 were gradually improved except AP_L. Ablation experiments proved that each module could improve the SCLANet.

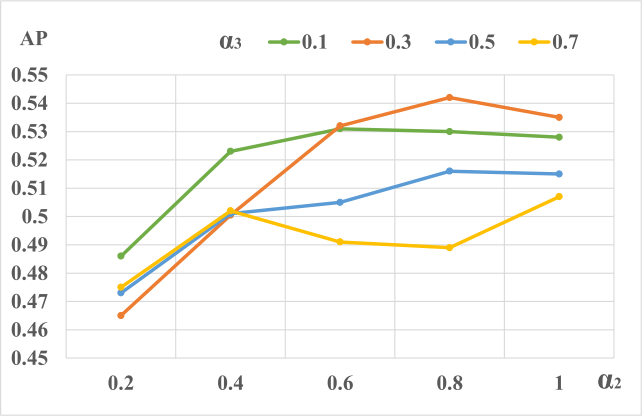


Fig. 9. Results of SCLANet with different parameters. Fix the value of α_1 to 1 and change the value of α_2 and α_3. When α_2 is equal to 0.8 and α_3 is equal to 0.3, the performance of SCLANet is the best.

In the training process, the values of parameters α_1, α_2, and α_3 in formula (13) would greatly affect the performance of the SCLANet. Parameter analysis experiments were carried out to compare the performance of the SCLANet under different parameter values. We fixed the value of α_1 to 1 and changed the value of α_2 and α_3. Fig. 9 and Table IX show the results of the parametric analysis experiment. Table IX shows the AP values calculated by SCLANet trained with different parameters. When α_3 equaled 0.1 and α_2 increased from 0.2 to 0.8, the AP value increased from 0.4862 to 0.5304. When α_3 equaled 0.1 and α_2 was 1, the AP value decreased to 0.5280. When α_3 equaled 0.3, the highest AP value was 0.5421, and the lowest was 0.4653. It is easy to see that the value of AP increased first and then decreased with increasing α_2. α_3 was equal to 0.5 and 0.7, which also occurs. In addition, when α_3 is equal to 0.3, the AP value was higher as a whole; when α_3 is equal to 0.7, the AP value was lower as a whole. Moreover,

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TABLE VI
RESULTS OF ALL ALGORITHMS WITH DIFFERENT PROPORTIONS OF LABELED DATA

		Faster R-CNN	YOLOv4	CSD-SSD	Unbiased-teacher	STAC	UFO2	SCLANet
5%	AP	0.3021	0.2814	0.3876	0.4089	0.4033	0.3954	**0.4195**
	AP_{50}	0.6376	0.5855	0.7168	0.7486	0.7318	0.7070	**0.7494**
	AP_{75}	0.2733	0.2521	0.3759	**0.3950**	0.3878	0.3702	0.3936
10%	AP	0.2928	0.2803	0.4058	0.4152	0.4098	0.4017	**0.4300**
	AP_{50}	0.6193	0.5934	0.7267	0.7314	0.7422	0.7096	**0.7560**
	AP_{75}	0.2868	0.2697	0.3791	0.3968	0.3825	0.3736	**0.4053**
15%	AP	0.3720	0.3324	0.4325	0.4613	0.4303	0.4286	**0.4641**
	AP_{50}	0.7004	0.6439	0.7436	0.7898	0.7661	0.7435	**0.7969**
	AP_{75}	0.3411	0.2793	0.4124	**0.4395**	0.4148	0.4098	0.4380
20%	AP	0.3934	0.3478	0.4554	0.4891	0.4677	0.4563	**0.5002**
	AP_{50}	0.7358	0.6629	0.7615	0.8130	0.7932	0.7621	**0.8412**
	AP_{75}	0.3599	0.2887	0.4471	0.4736	0.4603	0.4400	**0.4835**

TABLE VII
INFERENCE TIME

Method	Faster R-CNN	YOLOv4	CSD-SSD	Unbiased-teacher
Time/s	0.0460616	0.0353485	0.0208858	0.0460616
Method	STAC	UFO2	SCLANet	
Time/s	0.0460616	0.0460616	0.0462082	

TABLE VIII
AP OF DIFFERENT CASES

	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
Case 1	0.4595	0.7664	0.4283	0.4031	0.5328	0.4853
Case 2	0.4829	0.8077	0.4672	0.4316	0.5470	0.4371
Case 3	0.5235	0.8381	0.4962	0.4734	0.5908	0.4768
Case 4	0.5420	0.8825	0.5341	0.4847	0.6209	0.4226

TABLE IX
AP UNDER DIFFERENT PARAMETERS

		α_2				
		0.2	0.4	0.6	0.8	1.0
α_3	0.1	0.4862	0.5236	0.5310	0.5304	0.5280
	0.3	0.4653	0.5005	0.5322	0.5421	0.5353
	0.5	0.4735	0.5015	0.5054	0.5162	0.5153
	0.7	0.4752	0.5027	0.4917	0.4891	0.5074

it was not conducive to the training of the SCLANet when α_3 was more than 0.5. It was easy to conclude that, when α_2 is equal to 0.8 and α_3 is equal to 0.3, the performance of SCLANet is the best. In the above comparative experiment and ablation experiment, α_1 equals 1, α_2 equals 0.8, and α_3 equals 0.3.

V. CONCLUSION

In this study, we examined semisupervised learning in SAR images. The new SCLANet was proposed for semisupervised SAR ship detection. The SCLANet, which includes the adversarial learning module and the consistency learning module, was based on Faster R-CNN. Adversarial learning makes the local features extracted from unlabeled data closer to those extracted from labeled data. The consistency learning module consists of two parts: noise robustness consistency learning and output encoding consistency learning. They can be used as pretext tasks to train unlabeled data. Compared with other algorithms, the SCLANet showed better performance on two datasets. In different scenarios, SCLANet also showed better performance than other algorithms. In the experiments with different amounts of data, the result of the SCLANet was better than that of other algorithms. In addition, ablation experiments showed that each module was beneficial to the SCLANet. In future work, we will focus on semisupervised learning in various remote sensing applications. We want to apply transformer and semisupervised learning to SAR images. Based on the characteristics of SAR images, we wish to design better object detection algorithms for SAR images.

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